

Software Requirements Specification (SRS)

Project: Railway Reservation System

Version: 1.0

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Revision history

Version	Date	Author	Change summary	Approval
1.0	03-09-2026	Team 2	Initial SRS draft for Railway Ticket Reservation System	

Approvals

Role	Name	Signature / Email	Date

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Railway Reservation System (RRS). It defines functional and non-functional requirements, interfaces, verification criteria, and system architectural models intended for instructors, students, and evaluators to use as a formal engineering reference.

1.2 Scope

Covers passenger interactions at booking consoles and client terminals, train schedule and route inquiry, multi-class berth inventory tracking (1A, 2A, 3A, Sleeper, Chair Car), automated reservation with 10-digit PNR issuance, waitlist (RAC/WL) FIFO queue processing, ticket cancellation with dynamic slab refund deductions, operational counter management by clerks, and administrative master train catalog configuration. Excludes physical rail interlocking systems, signaling hardware, and third-party payment gateway banking internals beyond standard APIs.

1.3 Audience

Developers, QA Engineers, System Integrators, Station Booking Clerks, System Administrators, Course Instructors, and Assessment Evaluators.

1.4 Definitions

List of acronyms:

RRS: Railway Reservation System, **PNR**: Passenger Name Record, **CNF**: Confirmed Berth, **RAC**: Reservation Against Cancellation, **WL**: Waiting List, **FIFO**: First-In, First-Out queue scheduling, **CLI/TUI**: Command-Line Interface / Text User Interface, **RBAC**: Role-Based Access Control, **ACID**: Atomicity, Consistency, Isolation, Durability, **SHA-256**: Secure Hash Algorithm 256-bit

2. Overall description

2.1 Product perspective

The Railway Reservation System is a modular, high-performance console and networked reservation platform developed in C/C++. It serves as a central transaction engine handling concurrent multi-terminal booking requests. It integrates local binary/relational transactional persistence, thread-safe memory management, role-based user authentication, and deterministic state transitions across passenger booking lifecycles.

2.2 Major product functions

- Authenticate users across Passenger, Clerk, and Administrator roles with masked credentials.
- Train route and schedule inquiry by source, destination station codes, and journey dates.
- Real-time berth inventory checking across coach classes (1A, 2A, 3A, SL, CC) and quotas.
- Multi-passenger reservation with age-based fare concession calculation and automated berth allocation.
- Generation and tracking of unique 10-digit Passenger Name Records (PNRs).
- Ticket cancellation processing with slab-based penalty calculation and ledger refund processing.
- Automated FIFO queue promotion cascading RAC passengers to CNF and WL passengers to RAC upon cancellation.
- Administrative master management for onboarding new train routes, configuring coach compositions, and modifying fare-distance tariffs.
- Final passenger reservation charting and daily station revenue accounting.

2.3 User roles and characteristics

- **Passenger**: General public user; expects quick schedule searches, clear berth availability, transparent fare calculation, and prompt PNR status updates.
- **Booking Clerk/Counter Operator**: Station counter personnel; requires high-throughput data entry, bulk booking shortcuts, immediate ticket printing, and daily cash reconciliation.
- **Railway Administrator**: System supervisor; requires elevated access to maintain train timetables, adjust quota boundaries, inspect audit trails, and manage system backups.

2.4 Operating environment

- POSIX-compliant UNIX/Linux or Windows console environment.
- C/C++ runtime (GCC / G++ 9.0+).
- Terminal supporting ANSI escape codes and non-canonical password masking.
- Non-volatile disk storage for structured binary records or SQLite transactional store.

2.5 Constraints

- Memory safety: zero buffer overflows, memory leaks, or dangling pointers in C/C++ string and struct handling.
- Concurrency: thread-safe mutual exclusion (mutexes / file record locking) preventing race conditions and double-allocated berths during concurrent transactions.
- Transaction atomicity: ACID compliance ensuring power loss or process kill does not corrupt booking records.
- Standard compliance: reliance on portable standard libraries with decoupled business logic.

3. External interface requirements

3.1 User interfaces

Primary UI: Interactive ANSI-colored Command-Line Interface (CLI) / Text User Interface (TUI) with clear hierarchical menus. Accessibility: masked password inputs (*), high-contrast formatting, and explicit input validation error messages.

3.2 Hardware interfaces

- Standard keyboard input with stream buffer flushing.
- Monitor/terminal display (standard 80x24 and 120-column formatted grids).
- Local hard drive/SSD for persistent binary record storage.
- Optional receipt/ticket printer interface generating 40-column passenger receipts.

3.3 Software interfaces

- C/C++ Standard Runtime Library (<cstdio>, <iostream>, <vector>, <cstring>).
- POSIX thread subsystem (pthread / <mutex>) for multi-terminal thread safety.
- Binary file serialization subsystem managing fixed-size structs for records.
- Cryptographic hashing module executing salted SHA-256 digests.

3.4 Communications

- POSIX TCP/IP Sockets for client-server multi-counter communications.
- Binary packet or structured JSON over TCP with retry/backoff mechanisms.

4. System features

Each requirement below includes acceptance criteria and a reference test case. IDs follow RRS-F-####.

4.1 Authentication & User Management

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-001	Authenticate users using unique usernames and SHA-256 hashed passwords across Passenger, Clerk, and Admin roles.	Functional	High	Security / All Users	AC-RRS-F-001: Correct credentials grant role-specific menu access. Test: TC-Auth-01	Requires Crypto Module
RRS-F-002	Mask password entry and lock accounts after 3 consecutive failed login attempts.	Functional	High	Security / PCI-DSS	AC-RRS-F-002: Characters echoed as '*' or suppressed; account locked. Test: TC-Auth-02	Terminal I/O
RRS-F-003	Register new passengers capturing unique User ID, Name, Contact, Age, and Gender.	Functional	High	Passenger	AC-RRS-F-003: Passenger profile persisted in user master. Test: TC-Auth-03	User Data Store

4.2 Train Schedule & Route Inquiries

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-004	Search for active trains by Source Station, Destination Station, and Date of Journey.	Functional	High	Passenger / Clerk	AC-RRS-F-004: Returns all matching train routes. Test: TC-Inq-01	Train Catalog
RRS-F-005	Display intermediate halt stations, timetables, and cumulative route distances.	Functional	Medium	Passenger	AC-RRS-F-005: Formatted timetable rendered. Test: TC-Inq-02	Train Catalog
RRS-F-006	Check real-time seat availability across all travel classes (1A, 2A, 3A, SL, CC) and quotas.	Functional	High	Passenger / Clerk	AC-RRS-F-006: Returns exact available count or WL/RAC status. Test: TC-Inq-03	Inventory Engine

4.3 Ticket Reservation & PNR Generation

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-007	Support multi-passenger reservations (up to 6 passengers) capturing Name, Age, Gender, and Berth Preference.	Functional	High	Passenger / Clerk	AC-RRS-F-007: Validates fields and count boundaries (1-6). Test: TC-Book-01	Inventory Engine
RRS-F-008	Calculate ticket fare dynamically based on base tariff, distance, class multiplier, and age concessions.	Functional	High	Finance / Passenger	AC-RRS-F-008: Senior citizen receives 40% concession. Test: TC-Book-02	Tariff Matrix
RRS-F-009	Allocate seats according to availability hierarchy: Confirmed (CNF) -> RAC -> Waiting List (WL).	Functional	High	Operations	AC-RRS-F-009: Transitions from CNF to RAC to WL automatically. Test: TC-Book-03	Inventory Engine
RRS-F-010	Assign specific coach and berth numbers, honoring preferences when possible.	Functional	High	Passenger	AC-RRS-F-010: Assigns berths based on inventory rules. Test: TC-Book-04	Berth Allocator

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-011	Generate a globally unique 10-digit numeric Passenger Name Record (PNR).	Functional	High	System Core	AC-RRS-F-011: PNR generated using timestamp-counter hash. Test: TC-Book-05	PNR Generator
RRS-F-012	Commit booking transactions atomically using file locks to prevent double-booking.	Functional	High	Integrity / Ops	AC-RRS-F-012: Prevents race conditions during simultaneous bookings. Test: TC-Book-06	Mutex / File Lock
RRS-F-013	Generate electronic ticket receipts showing PNR, Train, Berths, Total Fare, and Timestamp.	Functional	Medium	Passenger / Clerk	AC-RRS-F-013: Summary exported as text ticket file. Test: TC-Book-07	Reporting Module

4.4 Ticket Cancellation & Refund Processing

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-014	Allow cancellation of individual or all passengers on an existing ticket using PNR.	Functional	High	Passenger / Clerk	AC-RRS-F-014: Status updated in persistent store. Test: TC-Canc-01	Booking Store
RRS-F-015	Compute cancellation deduction charges based on time window prior to train departure.	Functional	High	Finance / Rules	AC-RRS-F-015: Deductions adhere to 48hr, 25%, and 50% rules. Test: TC-Canc-02	Tariff Matrix
RRS-F-016	Automatically cascade seat promotions: RAC to CNF, and WL to RAC upon cancellation.	Functional	High	Operations	AC-RRS-F-016: Strict FIFO queue promotion executed atomically. Test: TC-Canc-03	Queue Cascade Engine

4.5 Administrative Master & Charting

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-017	Allow Administrators to add, update, and deactivate train routes and schedules.	Functional	High	Railway Admin	AC-RRS-F-017: Updates persist across restarts. Test: TC-Adm-01	Train Master File
RRS-F-018	Configure coach compositions, class quotas, and maximum waitlist caps per train.	Functional	High	Railway Admin	AC-RRS-F-018: Quota modified without corrupting bookings. Test: TC-Adm-02	Quota Config
RRS-F-019	Update the distance-tariff matrix and dynamic pricing surge multipliers.	Functional	Medium	Railway Admin	AC-RRS-F-019: Future bookings use updated tariffs. Test: TC-Adm-03	Tariff Matrix
RRS-F-020	Generate a final Passenger Reservation Chart (Manifest) 4 hours prior to departure.	Functional	High	Station Master	AC-RRS-F-020: Outputs manifest and locks cancellations. Test: TC-Chart-01	Reporting Engine
RRS-F-021	Export daily station accounting reports summarizing revenue.	Functional	Medium	Accounts / Clerk	AC-RRS-F-021: Generates report matching audit log. Test: TC-Rep-01	Ledger Module

5. Non-functional requirements

Req ID	Requirement (shall...)	Category	Priority	Acceptance criteria / Measurement
RRS-NF-001	Train search and availability queries shall return results in <= 150 ms.	Performance	High	95th percentile latency <= 150 ms under load. Test: TC-Perf-01
RRS-NF-002	Booking transactions shall complete in <= 500 ms.	Latency	High	Average transaction time <= 500 ms with disk sync. Test: TC-Perf-02
RRS-NF-003	Achieve 99.95% transaction durability and crash resiliency.	Reliability	High	Zero corrupted records upon SIGKILL simulation. Test: TC-Rel-01
RRS-NF-004	Support at least 50 concurrent booking clerk sessions.	Concurrency	High	Zero double-allocated berths in stress tests. Test: TC-Conc-01
RRS-NF-005	Adhere to ISO C++17 standards, compiling with zero warnings.	Portability	Medium	Clean pass under -Wall -Wextra -Wpedantic. Test: TC-Port-01
RRS-NF-006	Clerk shall be able to complete an end-to-end booking in under 45 seconds.	Usability	Medium	Workflow completion time <= 45 seconds. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

1. Confidentiality: Protect passenger credentials and personal contact information.
2. Data Integrity: Guarantee ticket and inventory records cannot be modified outside transactional routines.
3. Authenticity & Non-Repudiation: Enforce strict RBAC and maintain a tamper-evident audit log.
4. System Availability: Safeguard against buffer overflows and resource deadlock.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
RRS-SR-001	Store passwords as salted SHA-256 cryptographic digests.	Security	High	Plaintext never stored or logged. Test: TC-Sec-01
RRS-SR-002	Enforce Role-Based Access Control (RBAC).	Security	High	Unauthorized attempts return 'Access Denied'. Test: TC-Sec-02
RRS-SR-003	Enforce strict boundary checking on all string and numeric inputs.	Security	High	Prohibit 'gets' and unsafe 'strcpy'. Test: TC-Sec-03
RRS-SR-004	Maintain an append-only, tamper-evident transaction audit log.	Security	High	Rolling SHA-256 block checksums used. Test: TC-Sec-04
RRS-SR-005	Encode a 32-bit CRC verification checksum within each PNR.	Security	High	Detect and reject forged ticket lookups. Test: TC-Sec-05

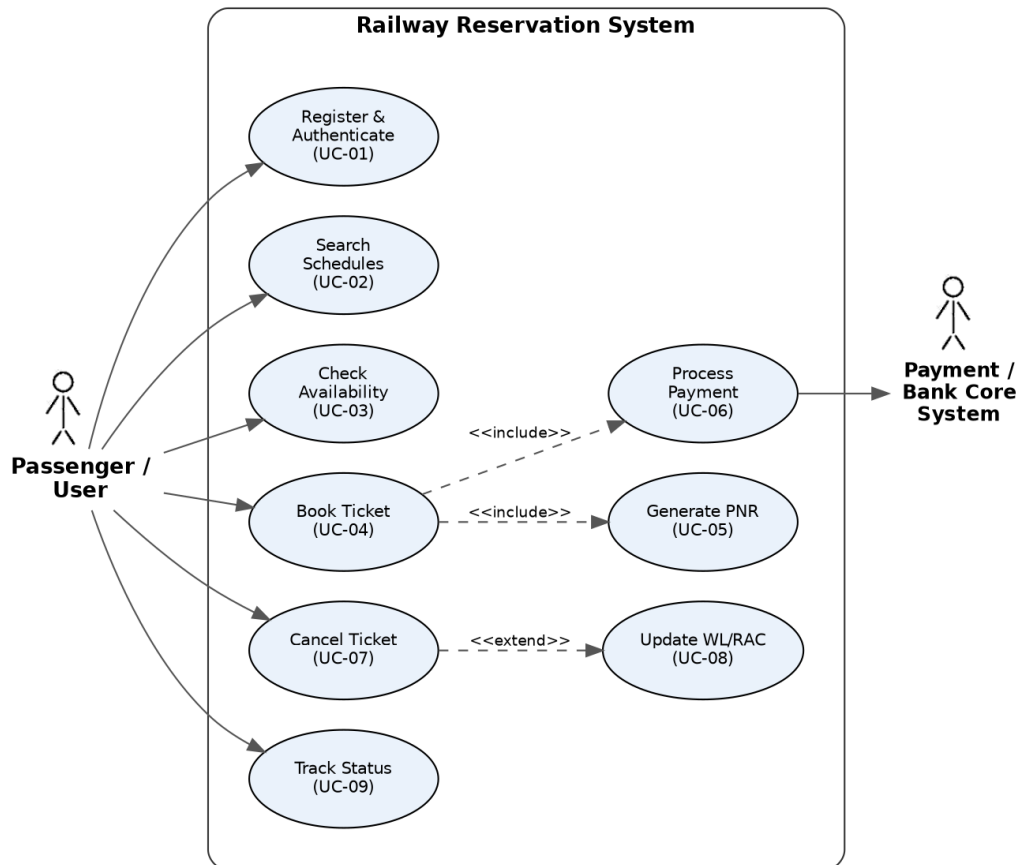
6. Quality attributes & Acceptance tests

- **Exit criteria for acceptance:** All 21 high-priority functional requirements implemented and verified, zero memory leaks reported under Valgrind, zero race conditions in concurrent stress testing, all 6 NFRs and 5 Security Requirements passed, and RTM confirms full test coverage.
- **Acceptance test suites:** Authentication & RBAC Suite, Train Search & Inquiry Suite, Booking & PNR Lifecycle Suite, Cancellation & Cascade Suite, Admin Master Suite, Security & Sanitization Suite, Performance & Concurrency Stress Suite.

7. System models and diagrams

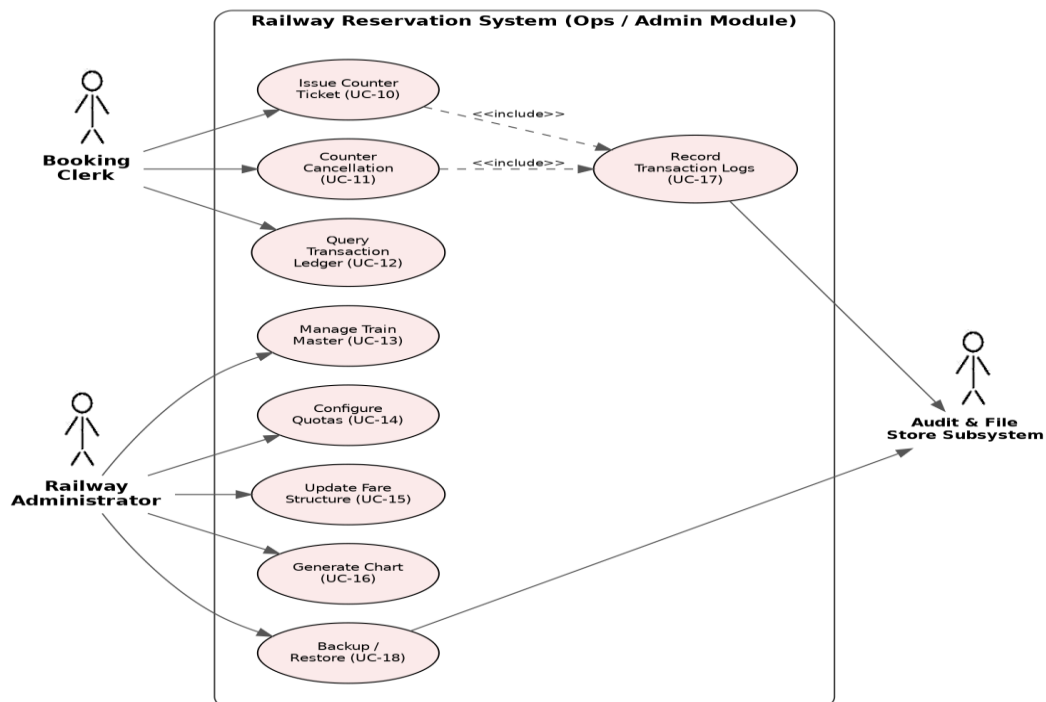
7.1 UML Use-Case Diagrams

Diagram 1: Passenger Ticket Booking & Cancellation Lifecycle



Actors: Passenger / User, Payment / Bank Core System. Booking (UC-04) includes Generate PNR (UC-05) and Process Payment (UC-06), which associates with the external Payment/Bank Core actor. Cancel Ticket (UC-07) extends to Update WL/RAC (UC-08) to trigger the FIFO promotion cascade.

Diagram 2: Railway Operations, Counter Clerk & Administration



Actors: Booking Clerk, Railway Administrator, Audit & File Store Subsystem. Issue Counter Ticket (UC-10) and Counter Cancellation (UC-11) both include Record Transaction Logs (UC-17), which writes to the Audit & File Store Subsystem, as does Backup/Restore (UC-18).

8. Requirements Traceability Matrix (RTM)

Status uses N = Not yet tested, P = In progress, A = Accepted/passed.

As this SRS precedes implementation, all items are marked N.

Req ID	Requirement (short)	Section ref	Module	Test case(s)	Status (N/P/A)	Comments
RRS-F-001	User Authentication	4.1	AuthModule	TC-Auth-01	N	
RRS-F-002	Password Masking	4.1	TerminalUI	TC-Auth-02	N	
RRS-F-003	Passenger Registration	4.1	UserStore	TC-Auth-03	N	
RRS-F-004	Train Schedule Search	4.2	CatalogModule	TC-Inq-01	N	
RRS-F-005	Intermediate Timetable	4.2	CatalogModule	TC-Inq-02	N	
RRS-F-006	Real-Time Availability	4.2	InventoryEngine	TC-Inq-03	N	
RRS-F-007	Multi-Passenger Form	4.3	BookingEngine	TC-Book-01	N	
RRS-F-008	Dynamic Tariff	4.3	TariffModule	TC-Book-02	N	
RRS-F-009	Seat Hierarchy	4.3	InventoryEngine	TC-Book-03	N	
RRS-F-010	Berth Allocation	4.3	BerthAllocator	TC-Book-04	N	
RRS-F-011	10-Digit PNR	4.3	PNRGenerator	TC-Book-05	N	
RRS-F-012	Atomic Transaction	4.3	TransactionStore	TC-Book-06	N	
RRS-F-013	E-Ticket Receipt	4.3	ReportingModule	TC-Book-07	N	
RRS-F-014	Cancellation	4.4	BookingEngine	TC-Canc-01	N	
RRS-F-015	Refund Slabs	4.4	TariffModule	TC-Canc-02	N	
RRS-F-016	Queue Cascade	4.4	CascadeEngine	TC-Canc-03	N	
RRS-F-017	Train Route Onboarding	4.5	CatalogModule	TC-Adm-01	N	
RRS-F-018	Quota Configuration	4.5	InventoryEngine	TC-Adm-02	N	
RRS-F-019	Pricing Updates	4.5	TariffModule	TC-Adm-03	N	
RRS-F-020	Manifest Charting	4.5	ReportingModule	TC-Chart-01	N	
RRS-F-021	Accounting Ledger	4.5	LedgerModule	TC-Rep-01	N	
RRS-NF-001	Search Latency	5	CatalogModule	TC-Perf-01	N	
RRS-NF-002	Booking Latency	5	BookingEngine	TC-Perf-02	N	
RRS-NF-003	Crash Durability	5	FileStore	TC-Rel-01	N	
RRS-NF-004	Concurrency Safety	5	TransactionStore	TC-Conc-01	N	
RRS-NF-005	Clean Compile	5	CoreEngine	TC-Port-01	N	
RRS-NF-006	Clerk Booking Time	5	TerminalUI	TC-UX-01	N	
RRS-SR-001	Salted SHA-256	5.1	CryptoModule	TC-Sec-01	N	
RRS-SR-002	RBAC	5.1	SecurityManager	TC-Sec-02	N	
RRS-SR-003	Buffer Protections	5.1	InputSanitizer	TC-Sec-03	N	

Req ID	Requirement (short)	Section ref	Module	Test case(s)	Status (N/P/A)	Comments
RRS-SR-004	Audit Log	5.1	AuditLogger	TC-Sec-04	N	
RRS-SR-005	PNR Checksum	5.1	PNRGenerator	TC-Sec-05	N	